Provability of A iff Gamma cup {not A} is inconsistent

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source follows this proposition with a problem asking for the dual statement Gamma proves (not A) iff Gamma cup {A} is inconsistent; that dual is the author's exercise and is not part of this card

Editorial note. The label prop:prov-incons names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Left negation rule. From Gamma sequent Delta, A one derives (not A), Gamma sequent Delta.. Forward direction: applying Left-not to the derivation of Gamma sequent A yields a derivation of (not A), Gamma sequent (empty).
- Cut and right negation. Right-not turns A sequent A into (empty) sequent A, (not A); cut with pi_1 gives Gamma sequent A.. Backward direction: the displayed derivation applies Right-not to the initial sequent A sequent A and cuts the result against pi_1.

Proposition 0.1. $\Gamma \vdash A$ *iff* $\Gamma \cup \{\neg A\}$ *is inconsistent.*

Proof. First suppose $\Gamma \vdash A$, i.e., there is derivation π_0 of $\Gamma \Rightarrow A$. By adding a rule, we obtain derivation of $\neg A, \Gamma \Rightarrow$, i.e., $\Gamma \cup \{\neg A\}$ is inconsistent.

If $\Gamma \cup \{\neg A\}$ is inconsistent, there is derivation π_1 of $\neg A, \Gamma \Rightarrow$. The following is derivation of $\Gamma \Rightarrow A$:

$$\dfrac{\dfrac{}{A A}\quad \begin{array}{c}\vdots\\ \pi_1\\ \vdots\end{array}}{\dfrac{A, \neg A \qquad \neg A, \Gamma}{\Gamma A}\,\text{Cut}}$$

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